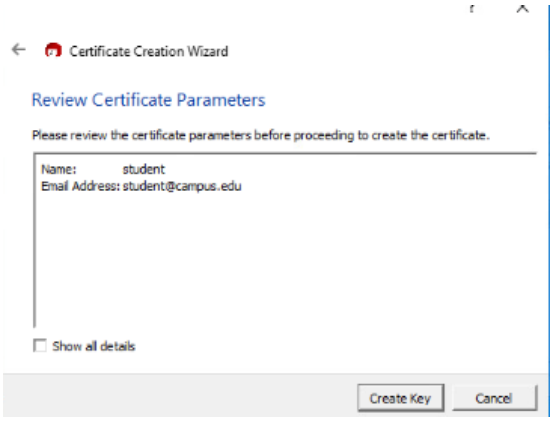
Lab 8.1: Using Public Key Encryption to Secure Messages

Lab 8.1: Part 1 – Creating the Certificate for Student

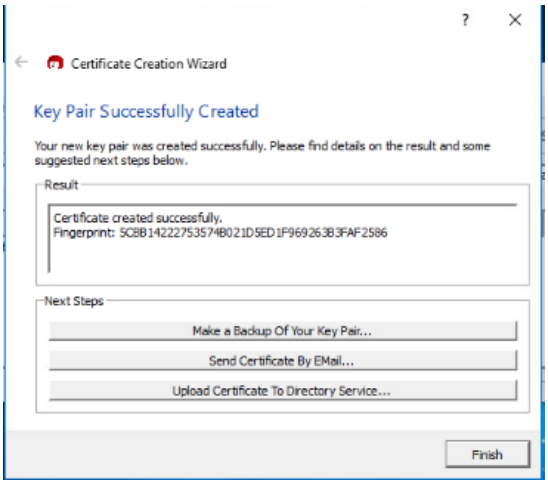
1. Research and describe what **Kleopatra** is

Kleopatra is a certificate manager GUI tool for GnuPG which is an email encryption standard

2. Question 5



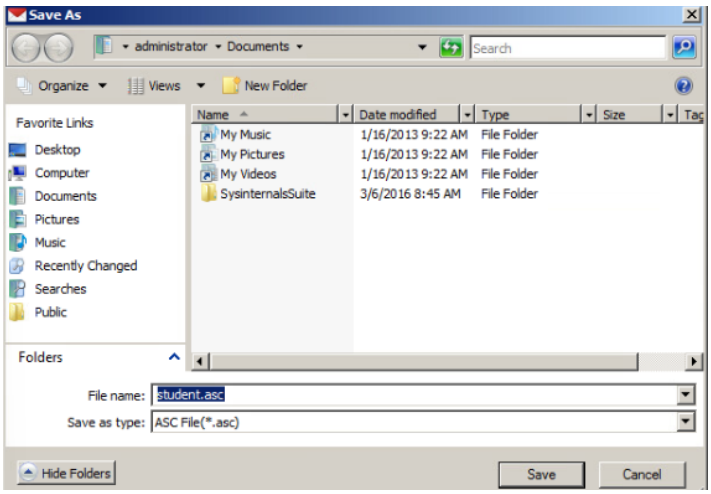
3. Question 10

4. 

5. Question 20

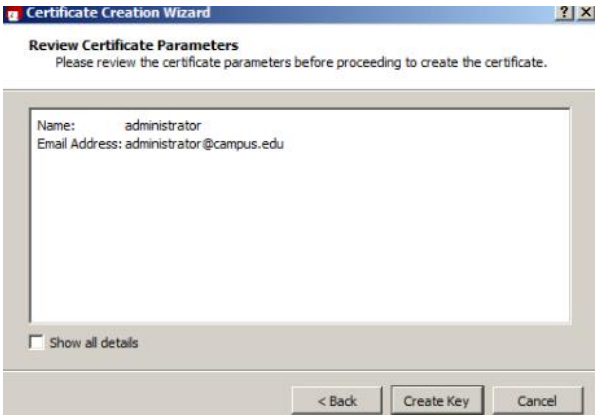
6.

7. Question 28

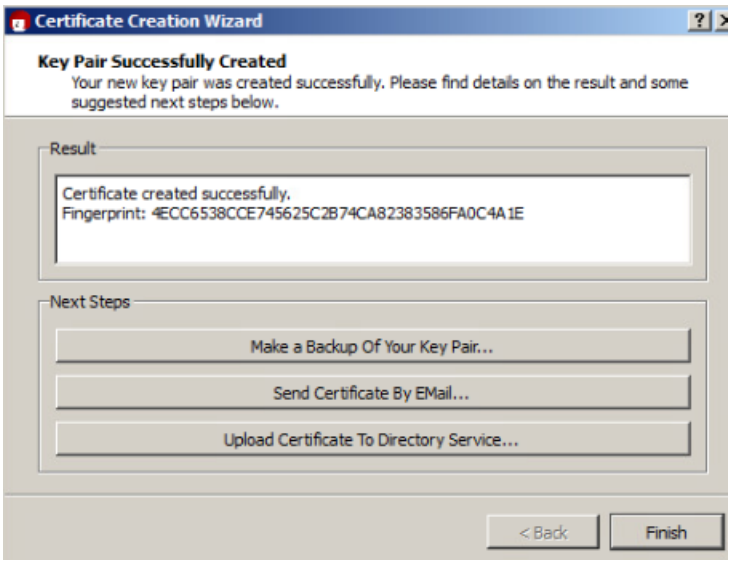


Lab 8.1: Part 2 – Creating the Certificate for Administrator

1. Question 5

2. 

3. Question 9

4. 

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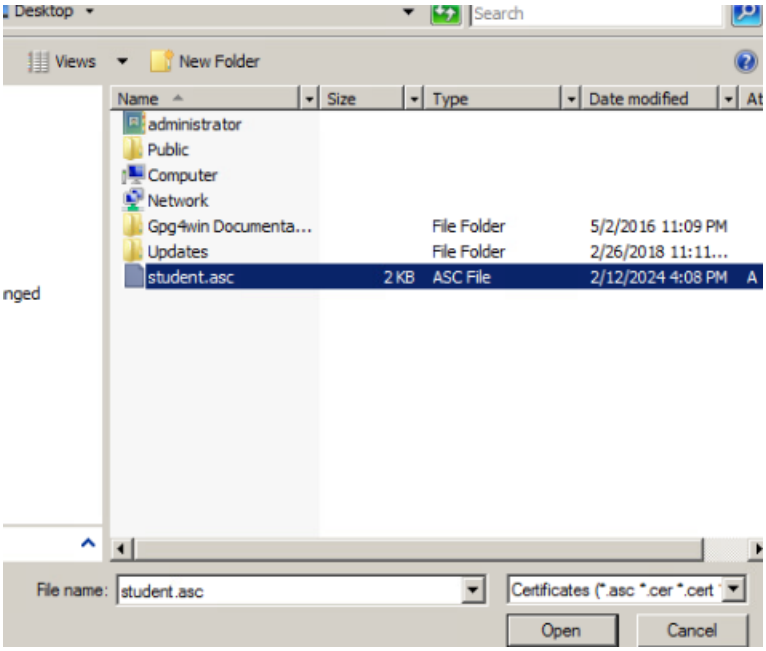
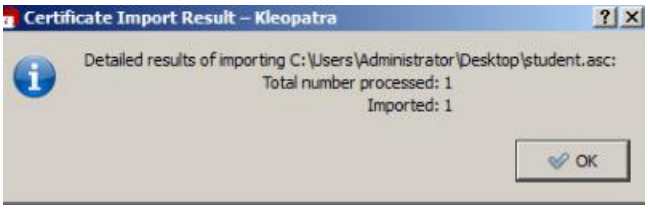
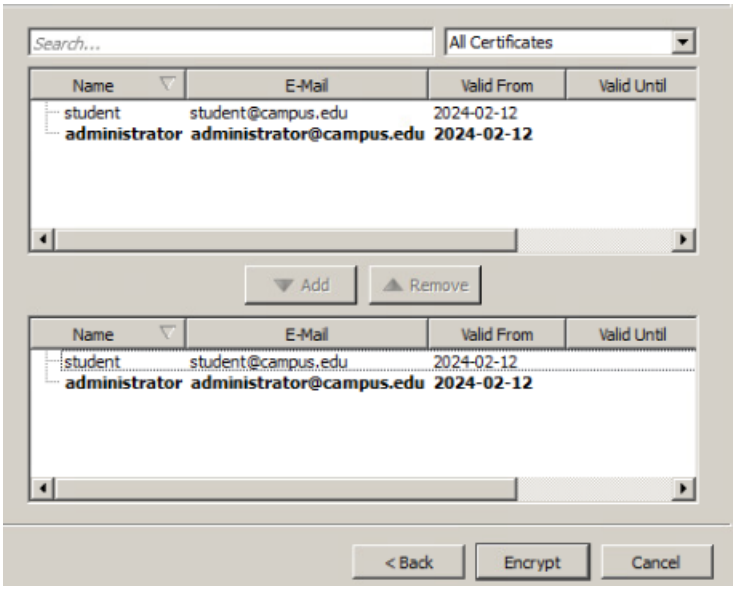
6. 

7. Question 13

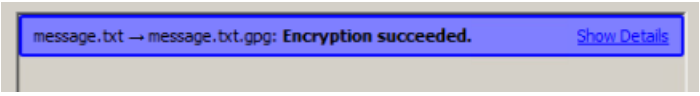


Lab 8.1: Part 3 – Encrypting and Decrypting the File

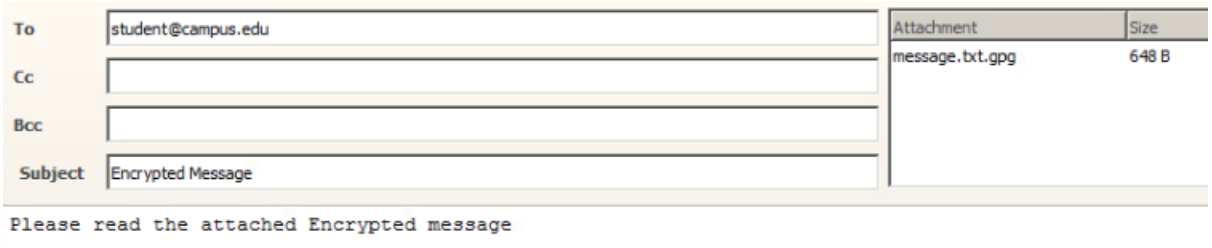
1. Question 10

2. 

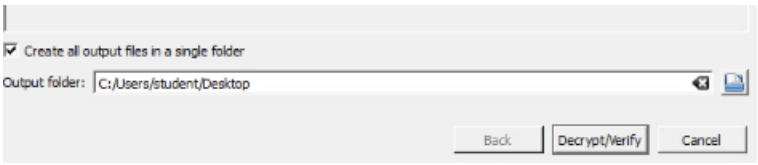
3. Question 11

4. 

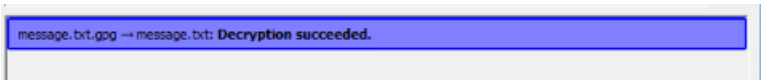
5. Question 16



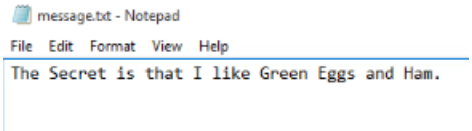
6. Question 24

7. 

8. Question 26

9. 

10. Question 28



Lab 8.1: Part 4 – Lessons Learned

Lessons Learned

Learned

- In this lab I learned a different use for encryption. The way I understand it is that you can have someone, or multiple people send you their certificate to make sure that it is going to a valid person. Then once you want to encrypt a file you can select whom you want to receive the encrypted file based on their certificate. You then can send that encrypted file through email and for anyone to view the encrypted file

they must have the password to the certificate it was sent to. I like this way of encryption because it makes it more secure in a way that each individual has their own passcodes to the encrypted messages being sent. Whereas if you were to encrypt a file with VeraCrypt you would need to send or tell the person the passcode which could get lost to the wrong person.

Surprises & Challenges

- The challenges I had with this lab is that it took a lot longer than it needed to complete the lab because the internet on campus has been super slow recently and while I was sending the email to the administrator, I accidentally put an extra u in campus because of the lag and had to figure out why I was not receiving the email.

Feedback

- This lab took me about an hour and a half to complete
- My favorite part was learning a new way to use encryption.